



TOSHKENT SHAHRIDAGI INHA UNIVERSITETI
INHA UNIVERSITY IN TASHKENT

School of Computers and Information Engineering

Spring semester 2025

Assignment #4 [Computer Networks]

Name:- _____ **ID:** _____ **Section:** _____

Submission Date: 13 /04/ 2025 [Sunday] (**23:30 Hr**)

NOTE :

1. Computer typed solution are not acceptable (**only handwritten** allowed)
2. Take the **print of this sheet** and solve on it.
3. Submissions not conforming to the given format will not be evaluated.

1. What is the size of an ARP packet when the protocol is IP and the hardware is Ethernet?

Answer

2. What is the size of an Ethernet frame carrying an ARP packet? Explain

Answer

3. What is the broadcast address for Ethernet? _____

4. A router with IP Address 125.45.23.12 and Ethernet physical Address 23:45:AB:4F:67:CD has received a packet for host destination with IP Address 125.11.78.10 and Ethernet physical Address AA:BB:A2:4F:67:CD (assume default values if needed)

- a. Draw the ARP packet with all its fields.
- b. Using reference of ARP packet in part (a). Draw ARP request packet sent by router for the above scenario and show all the entries in it. Assume no subnetting
- c. Draw separately and show how the packet made in part (b) is encapsulated in a data link frame. Fill all the fields

Answer